ZZ-H-421E
October 21, 1970
SUPERSEDING
Int. Fed. Spec. ZZ-H-00421D(GSA-FSS)
July 26, 1966 and
Fed. Spec. ZZ-H-421B
January 14, 1947

# FEDERAL SPECIFICATION

## HOSE, AND HOSE ASSEMBLY, CHEMICAL (INDUSTRIAL AND FIRE FIGHTING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hose and hose assembly used for conveying liquids of a chemical nature and for use with chemical fire fighting equipment.

1.2 Classification. Hose shall be of the following types, forms, classes, lengths, and sizes, as specified (see 6.2):

Type I - Chemical fire fighting.
Type II - Industrial chemical.

Form A - Braided.
Form B - Wrapped.

Class a - With couplings.
Class b - Without couplings.

1.2.1 Size. Hose shall be of the sizes and lengths shown in table I.

TABLE I. Sizes and lengths

| Size (inches) | Length (feet) |
|---|---|
| 3/8 | 1-1/2, 3 |
| 1/2 | 25, 50 |
| 5/8 | 25, 50, 100 |
| 3/4 | 50, 100 |
| 1 | 50, 100 |
| 1-1/4 | 50, 100 |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

WW-C-624 - Coupling Assembly, Hose (Fire, Garden, Water Suction).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 601 - Rubber; Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.



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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington. D.C., Atlanta, Ghicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-H-775 - Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of.
MIL-C-11053 - Coupling Assemblies, Hose (Chemical 3/4 and 1 inch).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc.. (UL) Standard:

UL92 - Chemical Engine Hose.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L. I., New York, 11746; 1655 Scott Boulevard, Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Type 1, forms A and B, chemical fire fighting hose.

3.1.1 Each contractor shall submit to the contracting agency proof that the hose he proposes to supply under this specification conforms to the requirements of UL 92. The label or listing of the Underwriters' Laboratories Inc., may be accepted as evidence that the hose conforms to this requirement (see 4.4.2).

3.1.2 In lieu of the label or listing, the contractor may submit independent proof satisfactory to the contracting agency that the hose conforms to the applicable requirements of the published standards including methods of test of UL 92.

3.2 Type II, forms A and B.

3.2.1 Material.

3.2.1.1 Rubber. The tube and cover shall be made of natural or synthetic rubber compound, or a mixture thereof, at the manufacturer's option.

3.2.1.2 Reinforcements.

3.2.1.2.1 Form A, braided. Plies of reinforcement shall be made of cotton yarn, rayon, or other suitable synthetic fibers, at the manufacturer's option.



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3.2.1.2.2 Form B, wrapped. Plies of reinforcements shall be made of cotton duck, rayon, or other suitable synthetic fibers, at the manufacturer's option.

3.3 Construction, type II.

3.3.1 Hose. Hose shall consist of a rubber tube, braided or wrapped fabric reinforcement, and a rubber cover.

3.3.1.1 Form A, braided. Braids shall be impregnated with the rubber compound, for joining the plies to the tube and cover. There shall be a distinct layer of the rubber compound between the plies of reinforcement. The number of plies shall be in accordance with table II.

3.3.1.2 Form B, wrapped. The wrapped fabric shall be evenly and firmly woven. The layers shall be applied with the edges lapped at least 1/2 inch and shall be impregnated with the rubber compound for joining the plies to the tube and cover. The number of plies shall be in accordance with table II.

3.4 Tolerances, type II, forms A and B.

3.4.1 Length. Tolerance for length shall be plus or minus 1.0 percent (see 4.4.3.1).

3.4.2 Thickness and diameter. Tolerances shall be in accordance with table II (see 4.4.3.2).

3.5 Couplings. Couplings shall be in accordance with either WW-C-624 or MIL-C-11053 as specified (see 6.2). The 3/8 inch size hose shall be furnished with couplings as specified (see 6.2).

3.5.1 Burst pressure. Couplings shall not separate from the hose when subjected to the test in 4.4.3.6.1.

3.5.2 Attachment of couplings. Female couplings shall be furnished with washers. Couplings shall be attached to the hose in a manner that will not injure the lining or cover.

3.6 Physical requirements, type II. Hose shall meet the requirements of table II (see 4.4.3).

3.7 Color, types I and II. Cover of hose shall be black or red, as specified (see 6.2). The shade of color shall be at the manufacturer's option.

TABLE II. Physical Requirements, type II

| Size, inside diameter, inches | 3/8 | 5/8<br>1/2 | 3/4 | 1 | 1-1/4 |
|---|---|---|---|---|---|
| Tolerance, inside diameter, plus or minus, inch | 1/32 | 1/32 | 1/32 | 1/16 | 1/16 |
| Outside diameter, maximum, inches | 13/16 | 1 | 1-5/16 | 1-5/8 | 1-7/8 |
| Thickness, minimum, inch: | | | | | |
| Tube | .060 | .060 | .060 | .060 | .060 |
| Cover | .047 | .047 | .047 | .047 | .047 |
| Reinforcement, total plies: | | | | | |
| Braided | 1 or 2 | 2 | 3 | 3 | 3 |
| Wrapped | 2 or 3 | 3 | 4 | 4 | 4 |
| Coupling and hose hydrostatic 1/<br>Test pressure, PSI minimum | 400 | 400 | 400 | 400 | 400 |
| Friction, minimum pounds: | | | | | |
| Between cover and plies | 10 | 10 | 10 | 10 | 10 |
| Between tube and plies | 10 | 10 | 10 | 10 | 10 |
| Between plies | 12 | 12 | 12 | 12 | 12 |



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TABLE II. (Continued)

| Size, inside diameter, inches | 3/8 | 5/8 1/2 | 3/4 | 1 | 1-1/4 |
|---|---|---|---|---|---|
| Tensile strength before aging, minimum pounds per square inch: | | | | | |
| Tube | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 |
| Cover | 1,200 | 1,200 | 1,200 | 1,200 | 1,200 |
| Ultimate elongation, before aging, minimum percent: | | | | | |
| Tube | 250 | 250 | 250 | 250 | 250 |
| Cover | 300 | 300 | 300 | 300 | 300 |

1/ The hydrostatic coupling test shall be made only when hose is purchased with couplings.

3.8 Accelerated aging, type II. Tensile strength and ultimate elongation of the tube and cover after accelerated aging (see 4.4.3.4) shall be not less than 60 percent of the tensile strength and elongation of the original values before aging.

3.9 Friction-type II. Friction shall be determined in accordence with 4.4.3.5.

3.10 Marking, types I and II. Each length of hose shall be indelibly marked with manufacturer's name, month and year of manufacture, and with the words "TESTED TO 400 PSI". Lengths of 50 feet shall have two markings per section approximately 25 feet apart. Lengths less than 50 feet shall have one marking per section; lengths over 50 feet shall have the markings spaced not more than 27 feet apart.

3.11 Workmanship, types I and II. Hose shall be uniform in thickness. The tube and cover shall be smooth, free from pits and blisters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of not more than 1500 feet of each type, class, and size hose or hose assembly submitted for inspection at the same time.

4.2.2 Visual examination of end item. Samples for visual examination shall be selected in accordance with level S-4.

4.2.3 Samples for tests. Samples for tests shall be level S-2.

4.3 Examination.

4.3.1 End item. Hose or assembly shall be examined for defects listed in table III. The Acceptable Quality Levels (AQL's) for defects shall be 2.5 percent defective for major defects, and 4.0 percent defective for minor defects.



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TABLE III. Classification of defects, end item

| Defects | Major | Minor |
|---|---|---|
| Type and class not as specified. | X | |
| Yarn not evenly and firmly braided. | X | |
| Fabric not evenly and firmly woven. | X | |
| Fabric sewed instead of lapped. | X | |
| Yarn or fabric not thoroughly saturated with rubber. | X | |
| Tube or cover not smooth. | X | |
| Tube or cover pitted. | X | |
| Tube or cover blistered. | X | |
| Incorrect size or length. | X | |
| Tolerances not within specified limits. | X | |
| Couplings improperly attached. | X | |
| Wrong color of hose. | | X |
| Marking illegible, improper or missing. | X | |
| Defects not affecting function or serviceability. | | X |

4.3.2 Inspection and preparation for delivery requirements.

4.3.2.1 Packaging test. To determine conformance to method of packaging specified in MIL-H-775, the package shall be subjected to the test specified in MIL-P-116 for the method of packaging used.

4.3.2.2 Packing test. To determine conformance to the method of packing specified in MIL-H-775, the level A packs shall be subjected to the interior packing test of MIL-STD-1186. Level B packs shall be subjected in the same test, except the height of the drop shall be 75 percent of the height of the drop specified herein.

4.4 Tests.

4.4.1 Acceptable quality level. The AQL for tests shall be 4.0 percent defective.

4.4.2 Type I. Chemical fire fighting hose shall be tested in accordance with UL 92 (see 3.1).

4.4.3 Type II. Industrial chemical hose.

4.4.3.1 Length (see 3.4.1).

4.4.3.1.1 Class a hose with couplings. The distance between the inside edges of the couplings shall be determined in accordance with method 2411, in Fed. Std. No. 601.

4.4.3.1.2 Class b hose without couplings. Length shall be determined in accordance with method 2411 in Fed. Std. No. 601.

4.4.3.2 Thickness, tube and cover. The minimum thickness of the tube and cover shall be determined in accordance with method 2011, in Fed. Std. No. 601 (see 3.4.2 and table II).

4.4.3.3 Tensile strength and elongation. These characteristics shall be determined in accordance with methods 4111, 4121, and 4131, in Fed. Std. No. 601 (see 3.6 and table II).

4.4.3.4 Accelerated aging. The resistance of the tube and cover shall be determined in accordance with method 7111 in Fed. Std. No. 601, except that the time of aging shall be 48 plus or minus 1/4 hour. Deterioration shall be measured by the change in tensile strength and elongation. Die VI in method 4111, shall be used for cutting the specimens (see 3.8 and table II).

4.4.3.5 Friction. Friction between cover and plies, tube and plies, and between plies shall be determined in accordance with method 8011, in Fed. Std. No. 601 (see 3.9 and table II).

4.4.3.6 Hydrostatic tests.

4.4.3.6.1 Class a, with couplings. Hose with couplings shall be tested in accordance with method 10211 in Fed. Std. No. 601, except that the hydrostatic pressure of 400 PSI shall be applied for two minutes.



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4.4.3.6.2 Class b, without couplings. Bursting strength of the hose shall be determined in accordance with method 10011 in Fed. Std. No. 601.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging shall be level A or C and packing shall be level A, B, or C as specified, (see 6.2).

5.1.1 Packaging and packing. Cleaning, packaging, and packing shall conform to Mil-H-775.

5.1.2 Marking.

5.1.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.1.2.2 Military activities. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-H-775.

6. NOTES

6.1 Intended use. Hose and hose assemblies are intended for conveying liquids of chemical nature, for use on fire extinguishers and fire fighting equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, form and class (see 1.2).
(c) Size and length (see 1.2.1).
(d) Type and size of coupling (see 3.5).
(e) Color of hose (see 3.7).
(f) Size of lot, if different from 4.2.1.
(g) Selection of applicable level of packaging and packing required (see 5.1).

MILITARY INTEREST:

Custodians:

Army - ME
Navy - YD
Air Force - 82

Review activities:

Army - WC
Navy - YD
Air Force - 82, 85

User activities:

Army - MI, MU
Navy - MC, CG

Preparing activity:

GSA-FSS

**CIVIL AGENCIES INTEREST:**

GSA
FAA
VA

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# SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, **and mail to General Services Administration, FSS (FMSK), Wash., D. C. 20406.**

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION ZZ-H-421e
HOSE, AND HOSE ASSEMBLY, CHEMICAL (INDUSTRIAL AND FIRE FIGHTING)

| 2. CONTRACT NO. *(If any)* | 3. QUANTITY ON CONTRACT *(Optional)* | 4. DOLLAR VALUE *(Optional)* |
|---|---|---|
| | | |

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

| 9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC. | 10. ADDRESS *(Number, Street, City, State and Zip Code)* |
|---|---|
| | |

| 11. NAME AND TITLE OF SUBMITTER | 12. DATE |
|---|---|
| | |

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